

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, a winding field and a uniform one are stationary at one coupling with opposite resultants, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together

with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are

21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 8.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand's relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator's stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable

44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
51 composes them, stating the thermodynamic constraints and all eight con-
52 necting hypotheses per branch; its supporting models of feedback, learning
53 and resource supply have distinct state spaces and assumptions. Section 6
54 then examines what the observables can establish, before Section 7 devel-
55 ops the cortical hypothesis and an awakening experiment. The hypothesis is
56 that extracellular contraction increases field-mediated coupling; its direction,
57 rate and relation to recovery require independent calibration. Supplemen-
58 tal Material organizes the constructions and controls behind the claims they
59 support.

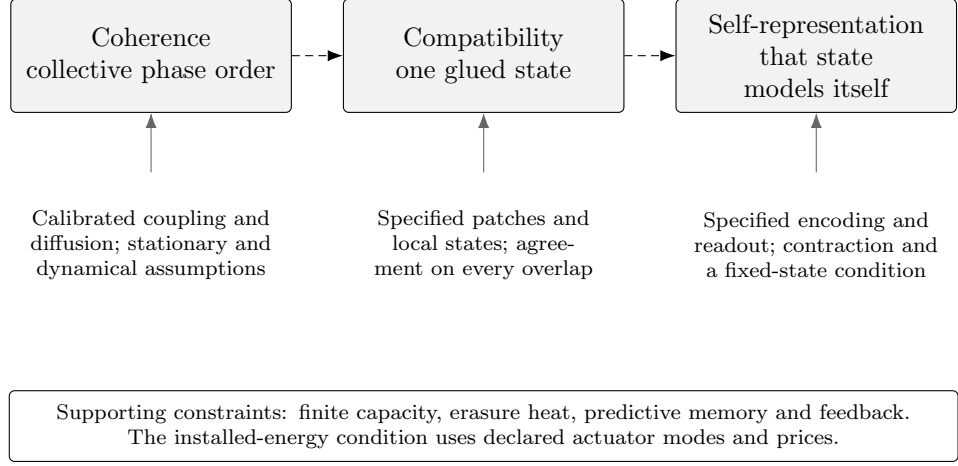


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

2 Coherence: physical coupling and its resource condition

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero.

83 These linear estimates concern the limiting equation. Relating them to finite
84 stochastic trajectories requires a dynamical mean-field limit and branch selec-
85 tion; exchange symmetry alone does not establish the needed independence
86 (Sznitman, 1991).

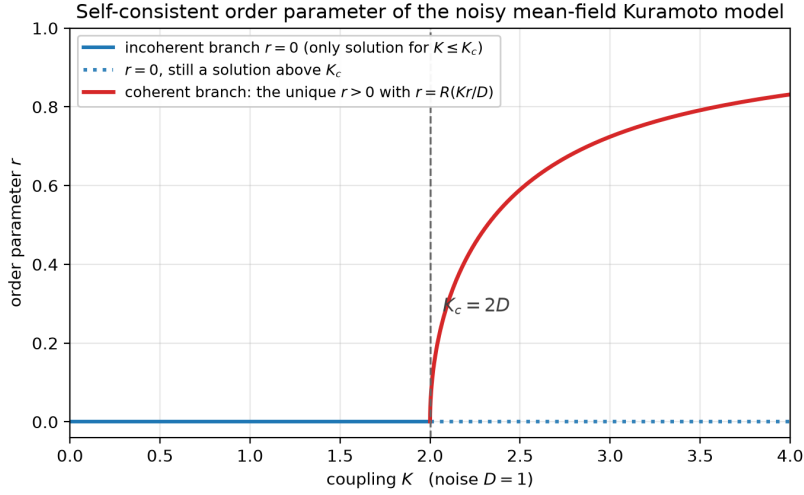


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

87 2.2 Installed energy bounds mean coupling

88 A spatial model replaces the finite sum by integration against a kernel. One
89 declared implementation carries response modes with continuous profiles ϕ_i ,

90 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

91 The same configuration sets the reciprocal kernel and stored energy. An
 92 executed transition changes the kernel through its changed occupancies and
 93 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 94 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 95 pair reconstruction on refining spatial partitions converges to its product-
 96 measure integral under the specified compactness and sampling assumptions;
 97 the refinement index is independent of the number of control updates.

98 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 99 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 100 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

101 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 102 coherent stationary branch and any growing Fourier mode about the uniform
 103 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 104 is necessary: energy above it need not produce sufficient coupling because
 105 the bound can be loose.

106 The quantity constrained is installed energy. Dissipated heat does not
 107 determine it, and κ is supplied by the mode profiles, prices and substrate
 108 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 109 tial dynamics requires a justified reduction. Identifying the installed kernel
 110 with a cortical field is a separate physical commitment.

111 **3 Compatibility: from coordinated activity** 112 **to a global description**

113 **3.1 Why common timing is insufficient**

114 For overlapping patches U and V , compatibility means that restricting either
 115 local description to $U \cap V$ gives the same object. Descriptions can agree on
 116 shared quantities while contributing different information elsewhere. Com-
 117 patible sections of the specified sheaf have a unique global extension. On
 118 a finite cover, local finite spatial measures agreeing as measures on every
 119 overlap also have a unique finite-measure extension whose sheaf image is the
 120 glued section. No mass-one normalization is imposed.

121 In the cup example, visual and reaching descriptions must express a
 122 shared quantity, such as cup location, in common coordinates. Restriction
 123 retains that quantity while dropping details particular to each description.
 124 Agreement permits surface appearance and hand configuration to contribute
 125 to one situation. The mathematical counterexample assigns the same phase

126 to every patch while two local mass profiles disagree on a shared site. Those
 127 profiles have no common extension. Because they are chosen independently
 128 of phase, synchrony places no constraint on their disagreement.

129 This distinction limits the biological interpretation. Receptive territories
 130 and content domains must be related to spatial patches, and the descriptions
 131 must support the required extension. Matching overlap marginals of de-
 132 coded probability laws need not determine a unique global joint distribution
 133 (Abramsky & Brandenburger, 2011).

134 **3.2 When coherence does constrain content**

135 A declared relation between phase and content gives a positive result. Let
 136 contents take values in a real normed vector space E . Patch i reads an
 137 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 138 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 139 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 140 sites; their equality on overlaps is a supplied relation between descriptions of
 141 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 142 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 143 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

144 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 145 agree exactly. The factor N makes this estimate potentially weak in large
 146 populations; it supplies a pointwise guarantee from a population average. A
 147 shared scalar encoder is a special case. Dropping the overlap relation permits
 148 disagreement even at perfect locking. A common nonexpansive linear projec-
 149 tion preserves agreement, but agreement of a projection does not establish
 150 agreement of the full vector.

151 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 152 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 153 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 154 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

155 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 156 duce an initial mismatch. Approximate observations leave a nondecaying
 157 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 158 update to the phase process.

159 **3.3 What approximate agreement determines**

160 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 161 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 162 Weights that vanish outside their patches and sum to one at each site select
 163 a global profile by weighted averaging. The selection lies within ε of each

164 patch there, and any two such selections differ by at most ε , independently
165 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
166 $L\varepsilon$ across selections.

167 At zero tolerance, the selection is the unique exact extension. At every
168 positive tolerance there are families with no exact extension; selection then
169 gives an approximation whose dependence on weights is bounded. Supple-
170 mental Material gives a three-site witness attaining the bound. The mass-
171 profile construction uses the scalar case of the content dynamics and addi-
172 tionally requires nonnegativity of the initial profiles and observations. The
173 general vector result supplies no identification with the spatial-measure sheaf.

174 These constructions concern declared objects: a specified cover, contents
175 valued in a normed space, and encoders related on overlaps. They select
176 neither the neural cover nor the decoder. Identifying their contents with
177 perceptual content, and establishing that the identification bears on experi-
178 ence, is a further modelling task; Section 6 sets out what a measurement of
179 it would have to establish.

180 4 Self-representation: a state that models it- 181 self

182 4.1 The reflexive construction

183 Compatibility gives a global state. Reflexivity relates it to an encoding within
184 that state. Equip global sections with a complete metric. If the additional
185 restriction-resonance condition identifies encoding with restriction to a region
186 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

187 The region and readout are fixed across the state space. If F contracts and
188 fixes the section s_* formed from the local descriptions, Banach's theorem
189 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
190 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
191 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
192 one for $\tau > 0$. Constructing a map with that factor remains a modelling
193 obligation.

194 In the cup example, the global state includes the represented scene and the
195 perceiver's relation to it. A local encoding must retain enough information
196 for the readout to recover that modelled state; cup position alone would be
197 insufficient if hand configuration and viewpoint could not be reconstructed.

198 For the same map with Lipschitz factor $q < 1$, approximate reconstruction

199 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1-q}. \quad (9)$$

200 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 201 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 202 weakens.

203 4.2 Reconstruction across a family of states

204 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 205 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 206 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

207 Thus M relevant states pairwise more than 2ε apart require at least M
 208 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 209 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 210 existing self-prediction map.

211 This is an additional accuracy criterion on a family, with no requirement
 212 for multiple exact fixed points. A blind encoder or constant readout fails it
 213 for separated states despite having a unique fixed point.

214 4.3 Agreement after a change takes communication

215 Snapshot compatibility does not specify how quickly descriptions respond
216 when a shared quantity changes. In a synchronous network whose updates
217 read only the site’s own state and declared neighbours’ messages, two initial
218 configurations agreeing on a site’s causal past through T rounds produce the
219 same state there at round T . If two fresh interventions outside that past
220 require reports separated by more than 2ε , no report from that site can be
221 within ε of both. This follows from the reconstruction bound in Section 4.2.

222 A three-site delay line witnesses failure after one round and exact report-
223 ing after two; cutting the path prevents delivery at every round. A report
224 may still happen to be correct about one fixed world. The bound concerns
225 responsiveness across interventions.

226 Throughout this section *prediction* names a reconstruction map; reading
227 it as temporal neural prediction would require dynamics. The relevant family
228 of states, the metric, the tolerance, the communication graph and the dead-
229 line are all declared inputs, as are the represented variables and the fidelity
230 counted as accurate. Reading the resulting code count as physical memory
231 capacity requires a hardware model, and a readout that varies with the state
232 requires a further one.

233 5 Thermodynamics and composition

234 The phase and representation results concern coupling, content disagreement
 235 and reconstruction error. Thermodynamics constrains the physical opera-
 236 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
 237 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-
 238 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
 239 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
 240 composition connects these objects through the explicit hypotheses in Ta-
 241 ble 1.

242 5.1 Feedback changes the information constraint

243 For passive memory X_t driven by an environmental signal S_t , the predictive
 244 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

245 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 246 et al., 2012). Zero retained information also satisfies it, so a task and a
 247 learning rule require additional specification. Relating this bound to Lan-
 248 dauer heat requires a common thermal scale and budget, and memory waste
 249 bounded by erased entropy.

250 An acting agent changes the signal it will receive. Specify policy, world,

251 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{12}$$

252 The second conditional term vanishes in the passive model. With feedback,
253 the signed difference can be negative as action creates correlations with the
254 future. A causal reading requires an environmental state sufficient to describe
255 the relevant dynamics.

256 For a controlled substep, the active thermodynamic bound instead uses
257 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
258 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
259 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
260 2013). Every forward path of positive probability must have a positive-
261 probability reverse for this finite identity. A missing reverse gives infinite
262 divergence. The supplement develops the support conditions and composes
263 actuation with memory updates on their shared intermediate law.

264 **5.2 Learning and repeated operation need explicit re-** 265 **sources**

266 The finite examples distinguish selecting a feasible policy, relaxing a register
267 toward an encoded reward, and learning an unknown task from observa-
268 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 2.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy E , thermal scale θ , local detailed balance and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

289 while continuing operation requires replenishment. Register–bath witnesses
 290 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply
 291 automatically.
 292
 293

294 5.3 What the composition establishes

295 The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named
 296 substep at the register’s thermal scale satisfies
 297

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

298 with B the register’s heat. The upper allocation follows from a ledger whose
 299 operations deliver that total heat to one reservoir and whose other operations
 300 do not increase joint entropy. Those ledger assumptions are physical inputs.
 301 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.

Table 1 (continued)

Connection	Status	Required assumption
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

303 lap compatibility and a reflexive boundary specifying encoding and readout.
304 They conclude `UnifiedSelf`: the glued section is the unique fixed point of
305 self-prediction. Compatibility and accuracy at that section enter as premises;
306 uniqueness follows. Identifying encoding with restriction needs the separate
307 resonance condition of Eq. (8).

308 Joint witnesses combine a register, boundary, refining mesh and small
309 spatial cover; active witnesses add controlled thermal updates and a priced
310 installation. They establish satisfiability under their stated identifications.
311 Mesh convergence follows from refinement data, and the witnesses establish
312 no common cortical mechanism. Supplemental Material maps the claims to
313 their Lean declarations and records the counterexamples. The development
314 declares no additional axioms beyond its libraries' foundational axioms.

315 6 What the observations establish

316 The composition holds under its premises. Whether observation can establish
317 those premises is a separate question, and the three requirements it connects
318 concern different objects: phase dynamics, local descriptions and internal
319 encodings. Their observables must retain these distinctions. The following
320 controls show which mechanisms remain indistinguishable, which temporal
321 approximations fail, which states one observable cannot separate, and what
322 independent content measurements must establish.

323 6.1 Stationary distributions leave the mechanism un- 324 determined

325 Equation (2) gives a concentration–coherence relation with no fitted shape
326 parameter. Estimate r from circular resultant length and a from density
327 shape: inverting R from the measured r would make agreement automatic.
328 Even independently estimated agreement identifies a statistical family shared
329 by different mechanisms.

330 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
331 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
332 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
333 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
334 and diffusion D_0 have that same density. Their stationary distributions co-
335 incide window by window despite different causes.

336 Resolved dynamics supply additional information. In the ideal model with
337 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
338 increment means constrain drift. Separating endogenous coupling from a
339 matched common drive additionally requires a measured drive or controlled
340 perturbation; identifying electromagnetic coupling requires evidence specific
341 to that interaction. Supplemental Material gives the equivalence construction
342 and its numerical check.

343 6.2 Finite ramps change the observable onset

344 The square-root law in Eq. (3) describes stationary solutions. A changing
345 coupling requires time for the phase distribution to approach those solutions.
346 In a deterministic mean-field control that crosses the threshold symmetrically,
347 none of the scanned rates meets the declared tracking tolerance: slowing
348 the crossing does not remove the terminal residual under that initialization.
349 A slow control variable therefore supplies no general guarantee of branch
350 tracking at onset. Supplemental Material gives the residual, rate range and
351 admissible rates away from threshold.

352 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
353 at four coupling speeds. Escape is the first of three consecutive saved samples
354 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
355 replicas escape within the fastest window. The three slower ramps have
356 mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents
357 differ from those of the exact stationary branch evaluated on the same grids.
358 At the fastest speed the ensemble mean never reaches the fit window, so its
359 fitted slope does not estimate stationary onset.

360 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
361 makes the apparent exponent depend on ramp speed and fitting window.
362 The specified deviation-based critical ramp speed is not bracketed by this
363 sweep. With fluctuating coupling, the relevant stationary comparison also
364 depends on rate: the mean coupling approximates order in the tested fast
365 regime, while averaging stationary order over coupling works for the tested

slow drives confined to one side of threshold. Slow crossing drives can satisfy
neither approximation. These are finite-horizon observations, with controls
and parameters in Supplemental Material.

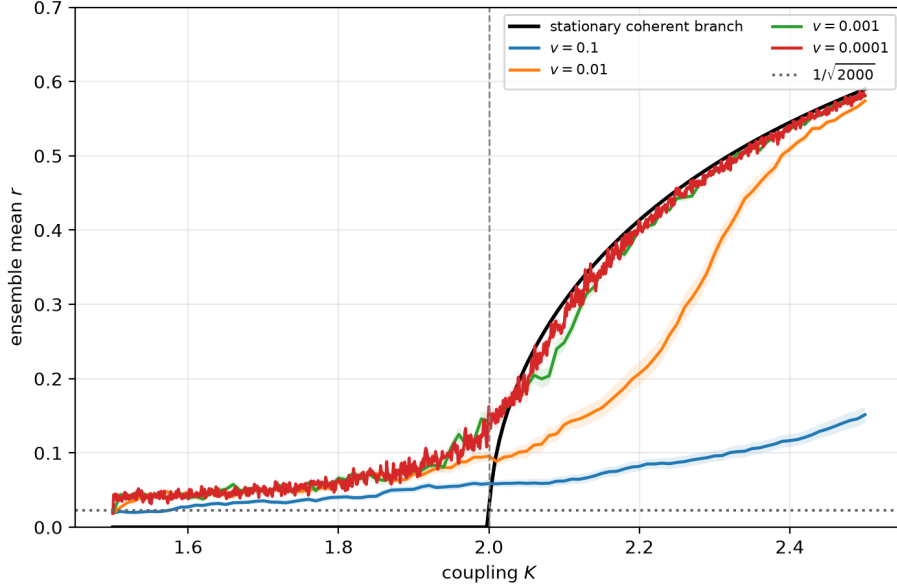


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

Additional controls address integration bias, population size and spatial
or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
rameters and controls in Supplemental Material; they supply no cortical cal-
ibration.

374 6.3 A winding state leaves the order parameter am- 375 biguous

376 The order parameter r is indifferent to where a phase sits in space. A field
377 winding once across a periodic sheet therefore has $r = 0$ while every patch
378 of it stays locked. Such a field is not an artificial construction here. On
379 a torus a twist is stationary for any isotropic symmetric kernel at identical
380 frequencies: the coupling term is odd in the separation against a kernel even
381 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
382 this for any finite periodic array, this sheet included. A twist satisfies the
383 positivity and identical-frequency hypotheses. It violates only the quarter-
384 turn initial spread that the convergence result additionally requires before
385 placing the limit at the uniform state.

386 On the sheet of Section 2's spatial control, a twist survives below an in-
387 terpolated decay length of 0.1871 mm and unwinds above it. That boundary
388 is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same
389 coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and
390 $r = 0.0083$ from a twisted one. Both are stationary, with patch-local order
391 0.9015 and no winding defect. At fixed K the observable therefore takes two
392 values spanning nearly its whole range. Coupling does not determine the
393 resultant, and a low resultant does not by itself distinguish weak coupling
394 from a winding field.

395 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

396 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 397 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realisations.
 398 Heterogeneity shortens the range that sustains a wave without removing it.

399 An amplitude layer decides which of these states exist, and leaves what
 400 the resultant reports unchanged. A phase state carries no amplitude. Every
 401 oscillator sits on the unit circle, so a field with an amplitude node has no rep-
 402 resentation in these dynamics. The real Stuart–Landau extension supplies
 403 one on the same coupling, and leaves intact the gradient structure the con-
 404 vergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\mu - |z_i|^2)z_i +$
 405 $\sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation with each cou-
 406 pling weighted by a ratio of amplitudes (`amplitude_phase_is_kuramoto`).
 407 The phase model is this one with the amplitude ratios set to one, rather
 408 than a separate object.

409 What the added degree of freedom decides is which windings exist at all.
 410 On a periodic array of any dimension, this sheet included, a winding requires
 411 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
 412 summed over separations, with ϕ_d the phase the winding advances across d
 413 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 414 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 415 lowers it further. A broad enough kernel therefore removes the winding out-
 416 right. The sweep above reaches that same boundary by a different route:
 417 the phase model retains every winding at every coupling, and loses this one
 418 under integration instead. The two layers bound the same regime from op-

419 posite sides. A phase recording does not separate a state the amplitude layer
420 excludes from one the dynamics fails to sustain.

421 These states are windings and not spirals, and on the sheet that is a
422 theorem rather than a shortage of dimensions. A cortical phase singularity
423 is a point at which the amplitude vanishes and around which the phase turns
424 once, read in practice as the circulation around a small plaquette. The states
425 above have equal modulus at every site, so that circulation is exactly zero on
426 every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can the
427 layer host a single defect. Over a periodic sheet the circulations of *any* phase
428 field sum to zero, so defects arrive in cancelling pairs (`no_isolated_defect`).
429 The sweep's near-zero measured defect density agrees with the first of these
430 and says nothing beyond it.

431 Two limits bound the layer for a reader of the cortical literature. Relax-
432 ational dynamics has no propagation speed, which is the quantity a cortical
433 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
434 sured envelope is not; this work identifies the two nowhere. Supplemental
435 Material gives the sweep, its uniform control, the amplitude layer and the
436 standing wave it admits inside the band, together with the remaining limits
437 of both.

438 This qualifies what a phase trace can establish. An onset criterion stated
439 in r alone registers a change of basin as readily as a change of coupling.
440 The protocol of Section 7.3 therefore needs a spatial observable beside the
441 resultant.

442 **6.4 Coherent adaptation can lose environmental align-** 443 **ment**

444 An adaptive coupling rule might supply representational structure, but its
445 objective must reward the relevant agreement. For symmetric coupling, pair-
446 wise sine terms cancel in the population mean drift. Minimizing squared drift
447 at that fixed mean therefore rewards equalizing drift across nodes. In the
448 frequency-cluster control, this favors coupling between unlike-frequency clus-
449 ters, whereas the diagnostic environmental template rewards coupling within
450 clusters.

451 The formalization proves descent of the squared-drift objective for a spec-
452 ified gradient flow at fixed phases. Its functional differs from thermodynamic
453 entropy production: at a coherent stationary density the probability current
454 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
455 rial gives the operator, its gradient and the conditions of the descent result.

456 The finite control co-evolves noisy phases and symmetric, nonnegative
457 couplings at fixed total resource. Its template is a diagnostic covariance of
458 latent signals, never a training input. Across three seeds, second-half order
459 remains between 0.9167 and 0.9222, while the within-cluster to between-
460 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
461 Random-update, relabelled-gradient and frozen-kernel controls are reported
462 in Supplemental Material.

463 The immediate post-update objective reduction disappears when aver-

464 aged over complete update intervals. A prespecified grid of 6 rate–cadence
465 combinations contains no case meeting the 5% sustained-reduction criterion
466 while retaining coherence and losing alignment. The finding is coherent adap-
467 tation away from the specified structure, with no demonstrated sustained cost
468 reduction.

469 Kernel alignment and overlap compatibility are different observables.
470 This control addresses acquisition of its designated template under its par-
471 ticular objective. Learning models can explicitly include data in an objective
472 (Corberi et al., 2026); an account of agreement about the cup must specify
473 the content its learning rule is meant to preserve.

474 **6.5 Content agreement needs independent decoders**

475 A neural test needs independently identified domains, local descriptions and
476 restriction maps. Two decoders should estimate a shared content on overlap-
477 ping domains before their agreement is examined. Discrepancy requires a null
478 under decoding noise and controls using deliberately compatible and incom-
479 patible examples. An implementation on constructed data separates these
480 cases at $AUC = 0.990$. Shared-prior shrinkage can nevertheless manufacture
481 agreement, and regional bias can manufacture disagreement. Per-patch de-
482 coding diagnostics detect these failures; the compatibility statistic alone does
483 not.

484 Phase agreement cannot replace that measurement. On constructed con-
485 tent, phase carries a share 1.000 of the content entropy when content is built

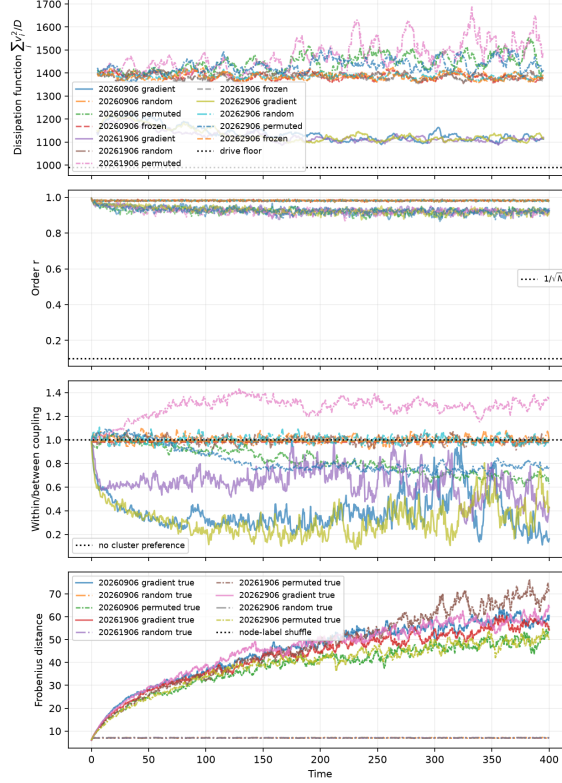


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

486 from it and 0.000 when it is not. A phase-derived statistic reports com-
487 patibility on phase-locked patches whose local descriptions disagree. The
488 implemented content statistic detects their incompatibility. These controls
489 validate a measurement on known constructions; a neural content model and
490 decoder remain to be established.

491 Choosing the cover also chooses a candidate subject: trivial descriptions
492 can agree while carrying little content, and a partition can conceal disagree-
493 ment. The useful choice must therefore be constrained by what neural ac-
494 tivity encodes. The exact composition assumes compatibility of its supplied
495 cover. The content mechanisms above identify additional hypotheses under
496 which agreement can arise; they leave the empirical choice of content and
497 the connection to that cover explicit.

498 **6.6 Phase-density estimation and measurement feasi-** 499 **bility**

500 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
501 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
502 itages produce different residuals. The bipolar results are compatible with the
503 Bessel relation, but their concentration range cannot distinguish it from its
504 linear approximation at the observed scatter. Temporal pooling changes the
505 estimand, and sedation recordings leave the awakening hypothesis untested.

506 The concentration range and dependence determine whether the relation

507 can be tested. Under the specified log-density estimator and bias-floor crite-
 508 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
 509 3000 independent sites in the scanned designs, rising to 31000 when sites
 510 share a mean in groups of 10. These counts describe that estimator and de-
 511 pendence model; the intended recording requires its own calibration. Supple-
 512 mental Material reports the bin-count controls and the full dependence-aware
 513 design.

514 An informative dataset needs sufficient concentration range or precision,
 515 calibrated estimators and an observation model for dependence, source mix-
 516 ing and reference choice. Public mouse calcium imaging provides thousands
 517 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
 518 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
 519 here. Site count, state coverage and bandwidth are separate requirements.

520 **7 The cortical hypothesis and an awakening**

521 **test**

522 The cortical proposal identifies a physical source of coupling and asks whether
 523 its changes help explain recovery. The tests in Section 6 determine what
 524 observations must distinguish before a phase trace can support this identifi-
 525 cation.

526 7.1 Physical identification and calibration

527 The cortical hypothesis identifies part of effective coupling with endogenous
528 extracellular electromagnetic interactions. Fields can modulate spike timing
529 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
530 propose coordination of distributed neural information (Pinotsis & Miller,
531 2022; McFadden, 2020). Phase organization under propofol motivates study-
532 ing this regime (Bastos et al., 2021), but does not measure coupling.

533 The formal field predicate requires a jointly continuous kernel, a normal-
534 ized substrate, coupling equal to the kernel average, and no single site pro-
535 viding more than half that strength. Toy constructions satisfy it. A cortical
536 realization must specify the scale at which the kernel and its reduction ap-
537 ply: calcium imaging reports different state-dependent modularity patterns
538 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

539 The spatial reduction is also testable. A row-sum aggregation at zero
540 diagonal tracks the uniform-coupling control over the tested spatial decay
541 range, while frequency heterogeneity displaces the operational threshold. The
542 fit window biases both estimates relative to $2D$; agreement with the control
543 is therefore the supported conclusion. Supplemental Material reports the
544 reduction, its numerical error and the distinct assumptions needed for the
545 kernel-average identification in Eq. (5).

546 The field commitment requires physical evidence beyond a choice of math-
547 ematical representation. A finite matrix spread over positive-mass cells pre-
548 serves its total coupling and phase correlation exactly. Point-supported

549 vanishing under an atomless measure and a disadvantage from excluding
 550 a better-correlated pair concern specified support constraints. Neither com-
 551 parison supplies energy prices or a biological–digital distinction.

552 A spike-time shift alone supplies no coupling rate. The dimensionless
 553 product of population size N , timing sensitivity s , field amplitude E and
 554 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

555 This rate describes how induced timing shifts accumulate into drift. Nei-
 556 ther it nor the installation factor κ follows from a heat bound. Testing the
 557 physical threshold requires independent estimates of the interaction kernel,
 558 its aggregation and diffusion in the same cortical regime.

559 **7.2 Extracellular geometry and recovery**

560 The proposed control variable is extracellular geometry. Extracellular vol-
 561 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 562 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 563 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 564 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 565 an effective-medium description predicts that a larger extracellular fraction
 566 raises conductivity and reduces field amplitude.

567 The coupling-gated hypothesis consequently predicts that extracellular

568 contraction on awakening raises E and, holding the other factors in Eq. (15)
 569 fixed, raises effective coupling. This argument supplies a direction; the
 570 geometry–coupling relation, its magnitude and its role in human recovery
 571 require measurement.

572 For a smooth crossing at nonzero speed, evaluating the stationary branch
 573 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

574 This follows from Eq. (3) at fixed D and concerns the stationary branch
 575 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 576 dently justified tracking approximation. The crossing controls in Section 6.2
 577 leave that approximation unsupported at onset, even though a separation of
 578 geometry and phase time scales can support it away from threshold. The
 579 awakening hypothesis therefore needs a calibrated dynamical prediction for
 580 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 581 does not supply a recovery latency.

582 **7.3 A protocol for the awakening experiment**

583 The physical test has six requirements:

- 584 1. **Coupling and diffusion in the same units.** Estimate them from re-
 585 solved phase dynamics, separating heterogeneity and observation noise.
 586 Measure the drive or perturb the system to distinguish endogenous cou-

587 pling.

588 **2. Geometry calibrated to coupling.** Validate the geometry observ-
589 able and its relation to effective coupling, with uncertainty. The con-
590 ductivity argument supplies only the expected direction.

591 **3. Calibrated phase measurements.** Estimate r by resultant length
592 and a by unweighted regression of log density on centered cosine.
593 Choose site count and binning from a dependence-aware calibration
594 over the intended concentration range. The tested designs require
595 thousands of sites at the observed concentrations; temporal pooling
596 changes the spatial estimand.

597 **4. An independent recovery endpoint.** Measure behavioral recovery
598 separately from phase organization.

599 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
600 old above the finite-size floor and evaluate its latency under the mea-
601 sured coupling trajectory, including ramp speed or fluctuations. Record
602 a spatial observable beside the resultant, since a winding field reports
603 a low resultant at coupling that sustains local order (Section 6.3).

604 **6. Matched model comparison.** Compare coupling increase, diffusion
605 change, common drive and alternative recovery models under the same
606 observation and filtering assumptions.

607 Failure of the concentration–coherence relation over an informative range
608 constrains the density model. Failure of the calibrated geometry–coupling
609 association constrains the physical gate. Better prediction of independent
610 recovery by a competing mechanism counts against its proposed role. A
611 favorable outcome still leaves the content of coordinated activity to be ex-
612 plained.

613 8 Discussion

614 Stated positively, and granting the premises collected in Table 1, the account
615 runs as follows. Extracellular contraction on awakening raises endogenous
616 field amplitude and with it the effective coupling between cortical oscillators.
617 Once that coupling passes $2D$, a coherent phase branch exists and the popu-
618 lation settles onto it. Encoders shared where cortical territories overlap turn
619 that phase order into agreement about the quantities the territories describe
620 in common, so the local descriptions glue: there is one state of which each
621 is a restriction. Part of that state encodes the state itself, and a readout of
622 that part reconstructs it accurately enough, and sensitively enough across
623 the states that matter, to make it the unique state the system models itself
624 as being. On this account the glued state is what makes distributed activity
625 one situation, and its accurate self-reconstruction is what gives that situation
626 a perspective.

627 Every step of that description is conditional, and its conditions are the

628 subject of what follows.

629 **8.1 Unity and a minimal self as empirical commit-** 630 **ments**

631 We propose the state glued from compatible local descriptions as a corre-
632 late of conscious unity, and its accurate internal reconstruction as a correlate
633 of a minimal self. The motivations are that different contents belong to
634 one experienced situation and that the situation is experienced from a per-
635 spective. The conditional results identify relations a realization would need:
636 sufficient installed resources for the specified phase model, agreement about
637 shared content, and an encoding that supports reconstruction at the required
638 fidelity.

639 Compatibility allows different descriptions to contribute different infor-
640 mation. It also permits uncertainty and conflicting judgments: apparent cup
641 position and judged cup position can be distinct quantities in a consistent
642 description of their discrepancy. Which quantities are shared must be de-
643 termined independently. Choosing them solely to secure agreement would
644 deprive the proposal of empirical force.

645 The minimal-self proposal concerns the perceiver’s position and condition
646 within the situation, such as the cup’s location relative to a body that can
647 reach it. The encoding could be implicit and need carry no autobiographi-
648 cal narrative. Its proposed necessity is philosophical: even an unreflective

649 episode has a perspective. Gluing by itself requires no self-encoding, and re-
650 constructability alone supplies no functional role. Altering the encoded hand
651 or viewpoint while holding the external scene fixed should change estimates
652 of the perceiver’s relation to the cup as the readout predicts. Whether that
653 represented perspective is experienced remains a further question.

654 **8.2 What a test of the full proposal must establish**

655 Both uniqueness results concern specified objects. They choose neither the
656 cover nor the encoding, and do not select between candidate subjects. In-
657 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
658 here selection requires independent evidence about neural organization. A
659 constant readout can contract and fix a predetermined state while discard-
660 ing its input. The family criterion in Section 4.2 tests the sensitivity that
661 this construction lacks, provided the relevant states and fidelity are declared
662 independently.

663 A test must separately measure coupling relative to diffusion, content
664 compatibility and self-reconstruction, alongside an independent endpoint for
665 experience. Meeting all three without a unified episode would challenge suffi-
666 ciency. A unified episode with incompatible descriptions would challenge the
667 unity identification; one lacking the specified self-encoding would challenge
668 the reflexive requirement. Awakening, dreaming and dissociative states offer
669 possible settings, but absence of report cannot by itself define absence of
670 experience.

671 The next physical requirement is a calibrated geometry–coupling relation
 672 and resolved phase dynamics. The installation bound contributes a nec-
 673 essary resource condition where its hardware representation applies; cortical
 674 mode prices and field-mediated coupling require measurement. For the repre-
 675 sentational connection, an informative experiment asks whether a controlled
 676 change in coordination alters independently decoded content agreement, with
 677 controls for sensory information and decoder quality. The constructed-data
 678 estimator establishes feasibility for specified contents. Neural local descrip-
 679 tions, shared overlap encoders and the self-encoding have yet to be identified.
 680 The population-size dependence of the phase-to-content bound further limits
 681 its quantitative use.

682 The separation results explain why these measurements must be inde-
 683 pendent. A stationary distribution leaves several recovery mechanisms indis-
 684 tinguishable. Equal phase order can accompany incompatible descriptions
 685 or different task information. The tested plasticity rule preserves coherence
 686 while losing its designated template. Each finding concerns the specified
 687 model and observable; their role is to delimit the evidence needed for the
 688 proposed connections.

689 **8.3 GPUs and the limits of no-go claims**

690 For a large language model (LLM) running on graphics processing units
 691 (GPUs), the framework requires identifying a physical system whose local
 692 descriptions form one state and whose internal encoding reconstructs that

693 same state. The coupling-gated route E56–E78 requires a calibrated physi-
694 cal coupling, its noise regime and a connection from coordination to content
695 agreement. An attention matrix or a shared clock supplies none of these iden-
696 tifications on its own. Computing a field model likewise leaves its realization
697 in the computing substrate to be established. These physical obligations also
698 apply to the cortical proposal.

699 Unity requires the candidate’s local descriptions to agree on indepen-
700 dently identified shared quantities (Section 3). Common timing and fluent
701 output do not establish that agreement. Digital representations can sat-
702 isfy compatibility, however, so finite hardware is compatible with the gluing
703 construction. The empirical commitment is to identify the relevant cover
704 and contents with one experienced situation; mathematical uniqueness alone
705 selects no subject.

706 For self, the system must carry an encoding $E(s)$ and an internal readout
707 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
708 the state produced by gluing. A constant readout shows why a unique fixed
709 point alone is insufficient for informative self-representation. The additional
710 family criterion in Section 4.2 supplies a test: if two relevant macrostates
711 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
712 within error ε . An intervention outside the encoding region’s causal past
713 gives a corresponding obstruction before a declared communication deadline.
714 Identical verbal reports alone do not establish identical internal codes. These
715 conditional no-go results exclude a specified reconstruction at a specified

716 fidelity or deadline. They apply to biological and digital candidates alike.

717 The hardware comparisons in Supplemental Material have similarly re-
718 stricted scope. Fixed wiring can impose a correlation disadvantage, and
719 point-supported kernels vanish under an atomless continuum measure. Nei-
720 ther comparison makes the representation conditions unattainable on GPUs.
721 A universal exclusion would need a necessary condition for consciousness and
722 a proof that every relevant implementation violates it. The present condi-
723 tional composition supplies neither.

724 Whether experience requires a living substrate is the explicit subject of
725 biological naturalism (Seth, 2026a,b). On that view the properties ground-
726 ing experience are those of self-sustaining organisms, and simulating them
727 does not secure them. The conditions stated here name no biological mecha-
728 nism. Computation described as such does not satisfy them either: a candi-
729 date must exhibit a physically realized coupling in a measured noise regime,
730 agreement about independently identified shared quantities, and an internal
731 reconstruction meeting a declared fidelity. Whether a realization of those
732 conditions that is not alive would be accompanied by experience is the point
733 at issue, and these results do not settle it.

734 The neuroscientific case against *human-like* consciousness in text-only
735 LLMs concerns embodied sensory interaction, thalamocortical organization
736 and biological regulation linking perception and action to survival (Aru et
737 al., 2023). Their absence supports skepticism, conditional on their causal
738 relevance to experience. Those mechanisms are not requirements of the ab-

739 stract gluing and reconstruction definitions. Workspace-like representations
740 reported in some LLMs (Gurnee et al., 2026) provide candidate organization
741 to investigate. Within this framework, the relevant evidence would identify
742 compatible contents and an informative internal reconstruction, or demon-
743 strate their failure under controlled interventions.

744 8.4 Explanatory scope

745 Integrated-information, global-workspace and predictive-processing accounts
746 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
747 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
748 candidate physical interaction and conditional relations between local content
749 and self-encoding. Gluing and reconstruction permit multiple substrates;
750 the electromagnetic commitment enters through physical calibration. The
751 formalization checks implications under their hypotheses, and its witnesses
752 establish realizability on declared models. They do not together supply a
753 cortical agent.

754 The explanatory question is whether the specified relations occur in neural
755 activity and matter for experience. In the cup example, this means establish-
756 ing how coordination helps maintain one represented situation and how the
757 perceiver’s place within it contributes to the episode. The proposal identi-
758 fies candidate correlates and tests of that identification. Why those relations
759 would be accompanied by experience remains unexplained.

760 **Data and code availability**

761 The Lean development, simulation scripts, tests and saved numerical sum-
762 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
763 `cortical_coherence`. Publication simulation values are generated from
764 saved summaries without rerunning production sweeps. The exploratory
765 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
766 processing and selection are described in Supplemental Material. The repos-
767 itory contains the reproducible analysis and does not redistribute the raw
768 recordings.

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770 This research received no external funding. The author declares no compet-
771 ing interests.

772 **Use of AI tools**

773 AI assistance to this work was substantial and is not separable by task. Sev-
774 eral coding assistants and chat interfaces were used in an interleaved way
775 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
776 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
777 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
778 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,

779 accessed with the Hermes CLI. Between them they contributed to the Lean
780 development, the simulation and analysis code, and the drafting and revision
781 of both documents. Because the tools were used in combination and in alter-
782 nation, no section, proof, figure or result is attributable to any one of them,
783 and none is claimed to be. The author is responsible for all content, including
784 everything a tool produced. Machine checking establishes the stated Lean
785 results under their hypotheses; it does not validate biological interpretations
786 or replace scientific review of generated code and prose.

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